TripCraft: A Benchmark for Spatio-Temporally Fine Grained Travel Planning

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Abstract

Existing travel-planning benchmarks rely on semi-synthetic data and evaluate itineraries with coarse binary metrics, leaving temporal scheduling, spatial feasibility, and personalization largely unmeasured. We introduce *TripCraft*, a realistic, transit-aware benchmark for spatio-temporally fine-grained itinerary generation. *TripCraft* contains 1,000 travel queries across 140 U.S. cities for 3-, 5-, and 7-day trips, each paired with a user persona and a human-annotated gold plan. The benchmark integrates General Transit Feed Specification (GTFS) public transit data, event calendars, and fine-grained attraction categories to enforce real-world constraints. Beyond conventional constraint pass rates, *TripCraft* introduces five continuous evaluation metrics: Temporal Meal Score ($\bar{T}_{meal}$), Temporal Attraction Score ($\bar{T}_{attrac}$), Spatial Score ($\bar{S}_{spatial}$), Ordering Score ($\bar{S}_{ord}$), and Persona Score ($\bar{S}_{persona}$). All metric parameters are inferred from human annotation statistics, ensuring alignment with expert-validated plans. We benchmark GPT-4o under two prompting regimes—standard and parameter-informed—and show that incorporating statistical priors substantially improves temporal scheduling while revealing a rigidity-versus-feasibility trade-off in constraint satisfaction. *TripCraft* sets a new standard for evaluating LLM-based travel planning in practical, personalized scenarios.

1 Introduction

Planning complex, multi-day itineraries is a quintessential task that requires reasoning over spatial, temporal, and personal constraints simultaneously. Large language models (LLMs) have demonstrated impressive capabilities in commonsense reasoning and tool use (Wei et al., 2022; Yao et al., 2023), making travel planning a compelling testbed for their practical utility (Xie et al., 2024; Xi et al., 2025).

Despite rapid progress in LLM-based agents (Xi et al., 2025; Song et al., 2023), existing travel-planning benchmarks suffer from three critical limitations. First, benchmarks such as TravelPlanner (Xie et al., 2024) rely on semi-synthetic or simplified data that omit real-world elements such as public transit schedules, event calendars, and fine-grained venue categories. Second, evaluation is largely binary: a plan either passes or fails each constraint, ignoring the degree to which a plan is temporally coherent, spatially convenient, or personally fitting. Third, persona-based personalization—a key factor in real-world trip satisfaction—is absent from prior benchmark designs.

These gaps motivate *TripCraft*, a benchmark specifically designed to probe the spatio-temporal fidelity and personalization quality of LLM-generated itineraries. *TripCraft* grounds every query in real-world, constraint-rich data: Points of Interest (PoIs) from OpenStreetMap and web scraping, GTFS-derived transit stop information for 140 U.S. cities, live event data, and millions of flight records. Each query includes a richly specified user persona and is paired with a human-annotated gold itinerary produced by trained graduate annotators.

Crucially, *TripCraft* replaces coarse binary pass rates with five continuous metrics whose parameters are calibrated from annotation statistics:

- **Temporal Meal Score** ($\bar{T}_{meal}$): measures how well meal timing and duration match human-annotated norms.
- **Temporal Attraction Score** ($\bar{T}_{attrac}$): captures alignment between attraction visit durations, pace preferences, and persona-driven attraction counts.
- **Spatial Score** ($\bar{S}_{spatial}$): rewards itineraries whose PoIs are close to public transit stops.
- **Ordering Score** ($\bar{S}_{ord}$): penalizes unnatural

orderings of accommodations, meals, and attractions via weighted edit distance.

- **Persona Score** ($\bar{S}_{persona}$): quantifies the semantic alignment between visited PoIs and the traveler's stated persona using BERT embeddings (Devlin et al., 2019).

We evaluate GPT-4o on *TripCraft* under two prompting conditions: a baseline setting without statistical priors (*w/o Parameter Info*) and an enhanced setting that injects annotation-derived timing and pacing statistics (*w/ Parameter Info*). Our experiments show that parameter-informed prompting substantially improves temporal quality (7-day $\bar{T}_{meal}$: $0.61 \rightarrow 0.80$) while exposing a revealing trade-off: constraint pass rates can decline when prompts push the model toward more realistic but harder-to-satisfy scheduling.

Our key contributions are:

- A realistic, transit-aware travel planning benchmark with 1,000 queries across 140 U.S. cities, spanning three trip lengths and three difficulty levels.
- Five novel continuous evaluation metrics that capture temporal, spatial, sequential, and personal dimensions of itinerary quality, with parameters derived from human annotation statistics.
- Human-annotated gold itineraries with per-decision rationales, enabling interpretable comparison.
- Empirical evidence that statistical priors improve objective itinerary quality while exposing a rigidity-versus-feasibility tension in constraint satisfaction.

2 Related Work

LLM Planning and Reasoning. The planning abilities of LLMs have attracted considerable attention (Valmeekam et al., 2023; Pallagani et al., 2023; Bohnet et al., 2024). Early work showed that LLMs can serve as few-shot planners for embodied agents (Song et al., 2023) and as commonsense heuristics in classical planning (Zhao et al., 2024; Borro et al., 2025). Prompting strategies such as Chain-of-Thought (Wei et al., 2022) and ReAct (Yao et al., 2023) further improve multi-step reasoning by interleaving thinking and tool use. More recently, Hao et al. (2025) demonstrated that LLMs combined with formal verifiers can achieve rigorous plan generation, while Lee et al. (2025) showed that evolutionary search at inference time yields near-perfect solutions on planning benchmarks. Despite these advances, most prior work focuses on constrained toy domains rather than real-world, personalized scenarios (Prasad et al., 2024; Huang et al., 2024).

Travel Planning Benchmarks. TravelPlanner (Xie et al., 2024) introduced a comprehensive benchmark for multi-day trip generation with nearly four million data records and 1,225 curated planning queries, evaluating constraint satisfaction via pass rates. Gundawar et al. (2024) demonstrated that the LLM-Modulo framework substantially improves TravelPlanner scores. ChinaTravel (Shao et al., 2024) extended this line of work to Chinese travel contexts with richer cultural constraints. However, these benchmarks share a common limitation: evaluation relies almost entirely on binary constraint checks, providing no signal about how well a plan schedules activities in time, situates them in space, or aligns with the traveler's preferences. *TripCraft* addresses these gaps with continuous, multi-dimensional metrics grounded in human annotation statistics.

Personalized and Transit-Aware Planning. Personalization in AI planning has been explored in general recommendation and task completion contexts (Singh et al., 2024), and travel agent systems have been proposed to handle user preferences (Chen et al., 2024). The integration of public transit data (GTFS) with LLM-based systems has been examined for urban mobility (Jonnala et al., 2025), but has not previously appeared in travel planning benchmarks. *TripCraft* unifies persona-conditioned query generation, GTFS-derived spatial constraints, and annotation-calibrated continuous metrics into a single evaluation framework.

3 *TripCraft* Benchmark

3.1 Overview

TripCraft is a travel planning benchmark designed to evaluate LLMs on realistic, persona-driven itinerary generation with spatio-temporal fidelity. Figure 1 illustrates the benchmark design. Each query specifies a departure city, destination(s), travel dates, group size, budget, and a detailed user persona. Models must produce a day-by-day itinerary that covers accommodations, meals, at-

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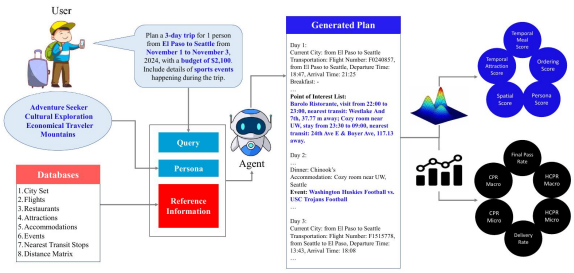

Figure 1: *TripCraft* overview: continuous evaluation metrics along with fine-grained itinerary details.

tractions, events, and transportation, and includes a timestamped Point of Interest (PoI) List with nearest transit stop information.

3.2 Dataset Construction

Data Sources. *TripCraft* integrates heterogeneous real-world data sources. PoIs (attractions and restaurants) were collected via web scraping and the OpenStreetMap (OSM) API, filtered to cities covered by GTFS public transit feeds. Accommodations were curated from listings sources. Events (sports, arts & theatre, music, and film) were scraped from event directories and synchronized to the trip date window. Flight data was adapted from TravelPlanner (Xie et al., 2024) with date alignment. For each PoI, the nearest GTFS transit stop and distance were computed and stored in a distance matrix, enabling spatial proximity scoring. Table 1 summarizes the scale of the resulting database.

Database	Data Entries (#)
City Set	140
Flights	3,446,829
Restaurants	3,892
Attractions	5,043
Accommodations	2,800
Events	21,980
Nearest Transit Stop	8,723
Distance Matrix	19,460

Table 1: Data entries in the database have been scraped to include the most recent data.

City Selection and Difficulty Levels. The 140 cities in *TripCraft* were selected based on GTFS data availability, ensuring that transit-based spatial scoring is meaningful. City coverage scales with trip length: 3-day itineraries visit 1 city, 5-day itineraries span 1–2 cities, and 7-day itineraries may cover up to 3 cities. Difficulty is determined by data availability: *easy* queries are in cities with abundant PoI and transit data, *medium* queries have moderate coverage, and *hard* queries involve cities with sparse data, requiring more careful constraint satisfaction. Table 2 shows the resulting dataset distribution.

Days	Easy	Medium	Hard	Total
3-day	130	106	108	344
5-day	87	164	73	324
7-day	99	178	55	332

Table 2: *TripCraft* dataset distribution

Query and Persona Generation. Each query was constructed by randomly sampling departure and destination cities, travel dates (aligned with available event data), group size, and budget. A detailed user persona was injected, comprising four components: *Traveler Type* (Laidback Traveler or Adventure Seeker), *Purpose of Travel* (e.g., relaxation, cultural exploration), *Spending Preference* (economical to luxury), and *Location Preference* (beaches, mountains, cities, forests). GPT-4o was used with few-shot prompting to generate natural-language query text from these structured components. Table 3 details the constraint and persona dimensions included in *TripCraft*.

Human Annotation Protocol. Gold itineraries were produced by 25 trained graduate annotators. Each annotator received the query, persona, and full database access, and produced a complete day-by-day plan with per-decision rationales explaining PoI choices and scheduling decisions. Each annotation required approximately 30 minutes and underwent multi-round review: an initial script-based constraint verification pass was followed by expert manual adjudication. Quality was enforced through iterative refinement cycles, ensuring geographic consistency, constraint compliance, and persona alignment across all plans.

3.3 Constraints

TripCraft enforces both commonsense and hard constraints. *Commonsense constraints* require a minimum four-hour gap between meals, valid PoI list formatting (all PoIs must appear in the database), and diverse event selections (no repeated event choices within a trip). *Hard constraints* specify the allowed event types (Sports, Arts & Theatre, Music, Film) and require that attractions belong

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Commonsense Constraints	
Sufficient Meal Gaps	Defines a minimum gap of four hours between the meals.
Valid PoI list	Defines validity rules for the Point of Interest list.
Diverse Events	Event choices should not be repeated throughout the trip.
Hard Constraints	
Event Types	Event Types include four distinct categories—Sports, Arts & Theatre, Music, and Film.
Attraction Types	Each attraction belongs to one or more of 15 predefined categories, ensuring a well-distributed selection of activities.
Persona Components	
Traveler Type	Defines how a traveler approaches their journey—whether they seek relaxation in cozy spots or adrenaline-pumping adventures.
Purpose of Travel	Captures trip motivation. Examples: to unwind, explore cultures etc.
Spending Preference	Reflects the traveler's budget and style, from luxurious indulgence to cost-conscious experiences.
Location Preference	Highlights preferred environments, such as beaches, mountains, cities, or wildlife-rich forests.

Table 3: Addition of constraints and persona details based on availability and limitations of scraped data has been given in the above table. The full list of constraints used in *TripCraft* has been given in Table 9 of Appendix.

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to one of 15 predefined categories, ensuring well-distributed activity selection. These constraints are verified using scripts adapted from TravelPlanner's evaluation framework (Xie et al., 2024), producing Delivery Rate, Commonsense Pass Rate (CPR), and Hard Constraint Pass Rate (HCPR) scores.

3.4 Continuous Evaluation Metrics

A key contribution of *TripCraft* is its suite of continuous metrics, whose parameters are inferred from the distribution of human-annotated gold plans. Figure 2 illustrates the motivation: two plans that appear identical under binary constraint checks may differ substantially in temporal scheduling, spatial convenience, and persona fit—differences that *TripCraft*'s metrics are designed to capture.

Temporal Meal Score ($\bar{T}_{meal}$). For each meal $m \in \{\text{breakfast, lunch, dinner}\}$, the temporal quality of a meal visit is measured by the normalized probability under a bivariate Gaussian over (start midpoint, duration):

$$T^{(m)} = \frac{\mathcal{N}([\tau_m, d_m];\ \boldsymbol{\mu}_m, \boldsymbol{\Sigma}_m)}{\max_{\boldsymbol{x}} \mathcal{N}(\boldsymbol{x};\ \boldsymbol{\mu}_m, \boldsymbol{\Sigma}_m)}, \tag{1}$$

where τ_m is the midpoint of the meal visit in hours, d_m is the meal duration, and $(\boldsymbol{\mu}_m, \boldsymbol{\Sigma}_m)$ are the mean vector and covariance matrix estimated from annotations. The overall temporal meal score $\bar{T}_{meal}$ averages $T^{(m)}$ over all meals across all days. Parameters for each trip length are given in Table 4.

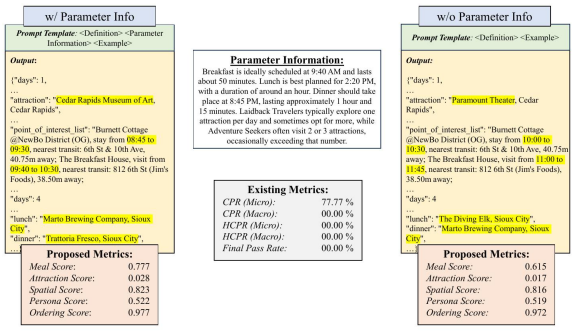

Figure 2: Existing constraint-based metrics rate these plans equally, but *TripCraft*'s metrics provide a continuous score capturing temporal, spatial, sequential, and personal differences.

Temporal Attraction Score ($\bar{T}_{attrac}$). The attraction score for a day combines the per-attraction duration score with a Poisson-distributed count model:

$$T_i^{(a)} = \exp\left(-\frac{(d_i - \mu_d^*)^2}{2\sigma_d^2}\right) \cdot \text{Poisson}(n;\ \lambda_p), \tag{2}$$

where d_i is the observed visit duration for attraction i, $\mu_d^* = \mu_{d,\text{type}} \pm k(n - n^{\text{ref}})$ adjusts the expected duration by the pace (λ_p-indexed) and count n, and $\lambda_p \in \{\lambda_{\text{laidback}}, \lambda_{\text{adventurous}}\}$ is selected based on the traveler type. The score $\bar{T}_{attrac}$ averages over all attraction visits. Category-specific duration priors $\mu_{d,\text{type}}$ are shown in Table 5.

Spatial Score ($\bar{S}_{spatial}$). The spatial score measures how close each PoI is to its nearest public transit stop, using a linear-exponential decay:

$$s(d) = \begin{cases} 1 - 0.5\,\frac{d}{5000} & d \le 5000\,\text{m}, \\ 0.5\,e^{-0.0002(d-5000)} & d > 5000\,\text{m}, \end{cases} \tag{3}$$

where d is the distance in meters. $\bar{S}_{spatial}$ is the average of $s(d)$ over all PoIs and days.

Ordering Score ($\bar{S}_{ord}$). Each day's PoI sequence is encoded as a type string over {x = accommodation, y = restaurant, z = attraction}. The Ordering Score compares the generated sequence against the gold sequence using Weighted Edit Distance (WED) (Levenshtein, 1965):

$$\bar{S}_{ord} = 1 - \frac{\text{WED}(\hat{s}, s^*)}{\max(|\hat{s}|, |s^*|)}, \tag{4}$$

averaged across days. A score of 1 indicates a perfect match with the annotated PoI ordering.

Persona Score ($\bar{S}_{persona}$). The persona score measures semantic alignment between visited PoIs and the traveler's stated persona using BERT embeddings (Devlin et al., 2019):

$$\bar{S}_{persona} = \frac{1}{|P|}\frac{1}{N}\sum_{j=1}^{|P|}\sum_{i=1}^{N}\cos(\mathbf{e}_{p_j}, \mathbf{e}_{v_i}), \tag{5}$$

where $\mathbf{e}_{p_j}$ is the BERT embedding of persona component j (Traveler Type, Purpose of Travel, Spending Preference, Location Preference), $\mathbf{e}_{v_i}$ is the embedding of PoI name i, and N is the total number of PoIs across all days.

3.5 Parameter Calibration

All metric parameters are inferred from the distribution of human-annotated gold plans, stratified by trip length and traveler type. Table 4 reports the key parameters for a subset of metrics. The Poisson rate $\lambda_{\text{laidback}}$ and $\lambda_{\text{adventurous}}$ capture the expected daily attraction counts for each traveler archetype, while the meal timing means and standard deviations reflect annotator consensus on natural meal schedules. These parameters are re-estimated separately for 3-, 5-, and 7-day trips to account for the different pacing typical of each duration.

4 Experiments and Results

4.1 Experimental Setup

We evaluate GPT-4o on *TripCraft* under two prompting settings. The *w/o Parameter Info* baseline uses the prompt template from TravelPlanner (Xie et al., 2024), providing the model with all retrieved database information, the query, and the persona, but no explicit statistical guidance on timing or pacing. The *w/ Parameter Info* setting augments this prompt with annotation-derived statistics: meal timing targets, meal duration priors, and persona-based attraction count recommendations (e.g., "Laidback Travelers typically explore one attraction per day; Adventure Seekers often visit 2 or 3 attractions"). Figure 3 shows the full data construction and evaluation pipeline.

Both settings use the same *sole-planning* approach (Xie et al., 2024), where all relevant tool outputs are pre-fetched and provided in a single prompt. We evaluate 1,000 queries (344 three-day, 324 five-day, 332 seven-day) across all difficulty

Parameter	3-day	5-day	7-day
Restaurant Parameters			
Breakfast			
Mean Duration (hrs)	0.90	1.08	0.85
Std. Duration (hrs)	0.24	1.43	0.23
Lunch			
Mean Duration (hrs)	1.11	1.10	0.99
Std. Duration (hrs)	0.36	0.35	0.26
Dinner			
Mean Duration (hrs)	1.19	1.32	1.15
Std. Duration (hrs)	0.43	0.91	1.15
Attraction Parameters			
$\lambda_{laidback}$	1.10	1.26	1.11
$\lambda_{adventurous}$	2.01	1.61	1.82
σ_d (hrs)	1.11	1.07	0.90
n^{max}	5	4	4
n^{min}	0	0	0
k (hrs)	0.28	0.28	0.28

Table 4: Day-wise parameter details calculated from our annotation distribution. For a complete list of parameters refer to the Appendix.

Category	Duration (hrs)
Boat Tours & Water Sports	3.5
Casinos & Gambling	2.5
Museums	3.0
Nature & Parks	4.5
Nightlife	2.5
Sights & Landmarks	3.0

Table 5: Attraction visiting duration (hrs) for a subset of categories (due to page limit). The complete list of categories has been given in Table 8 of Appendix.

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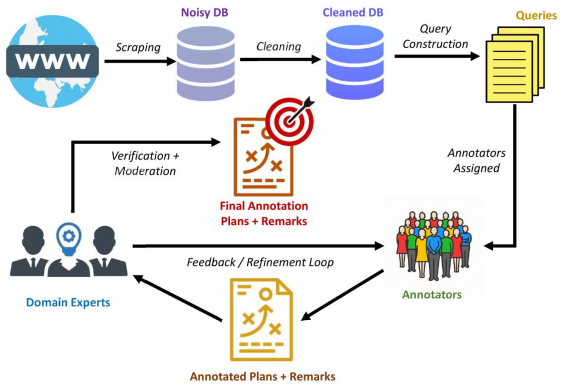

Figure 3: Visualization of the *TripCraft* construction and evaluation pipeline.

levels. Evaluation uses both the traditional constraint metrics (Table 7) and the five new continuous metrics (Table 6).

4.2 Results: Proposed Continuous Metrics

Table 6 reports results on the five proposed metrics. Incorporating parameter information yields consistent, substantial improvements in temporal meal scheduling across all trip lengths: the 3-day $\bar{T}_{meal}$ improves from 0.56 to 0.70 (+25%), 5-day from 0.69 to 0.75 (+9%), and 7-day from 0.61 to 0.80 (+31%). This demonstrates that explicit statistical priors effectively guide GPT-4o to schedule meals at times and durations closer to human-annotated norms.

Spatial convenience ($\bar{S}_{spatial}$) and PoI ordering ($\bar{S}_{ord}$) also improve consistently across all trip lengths: the 3-day spatial score increases from 0.80 to 0.83 and ordering from 0.70 to 0.74; 5-day ordering from 0.91 to 0.93; and 7-day ordering from 0.96 to 0.97. These gains suggest that pacing-aware prompting indirectly encourages more transit-proximate PoI selection and more natural sequencing.

Persona alignment ($\bar{S}_{persona}$) shows modest improvements (e.g., 3-day: $0.46 \rightarrow 0.48$, 7-day: $0.50 \rightarrow 0.51$), indicating that while persona-driven scheduling is partially improved by parameter priors, aligning PoI selection with persona components remains a challenging open problem.

The Temporal Attraction Score ($\bar{T}_{attrac}$) exhibits mixed behavior: 3-day improves marginally ($0.0169 \rightarrow 0.0171$), while 5-day improves ($0.0057 \rightarrow 0.0078$) but 7-day declines ($0.0122 \rightarrow 0.0057$). The low absolute values of $\bar{T}_{attrac}$ across all settings reflect the inherent difficulty of matching both visit duration and attraction count simultaneously under the joint Poisson-Gaussian model, suggesting room for further improvement in attraction scheduling.

4.3 Results: Existing Constraint Metrics

Table 7 reports Delivery Rate, Commonsense Pass Rate (CPR), and Hard Constraint Pass Rate (HCPR) for both settings. A clear trade-off emerges: while parameter-informed prompting improves several constraint metrics for 3-day trips (Delivery Rate: $92.60 \rightarrow 96.08$; CPR micro: $77.61 \rightarrow 80.08$), it degrades constraint satisfaction for longer trips. For 5-day itineraries, Delivery Rate drops from 98.69 to 91.30 and CPR micro from 66.99 to 61.69. For 7-day itineraries, Delivery Rate falls from 96.03 to 92.51.

This trade-off reflects a fundamental tension: prompts that enforce specific meal timing and pacing norms constrain the solution space, occasionally making it harder for the model to satisfy all hard constraints simultaneously. When meal slots are tightly scheduled and attraction counts are regulated by persona-based priors, the model has less flexibility to route around hard constraint violations. Across all settings, Final Pass Rates (plans satisfying all constraints simultaneously) remain very low (0–3.47%), consistent with findings from prior work (Xie et al., 2024; Gundawar et al., 2024) and highlighting the difficulty of *TripCraft* as a benchmark.

4.4 Discussion

Our results reveal several important insights. First, **continuous metrics are more discriminative than binary pass rates**: the parameter-informed setting is clearly superior on temporal and spatial quality despite mixed results on binary constraint satisfaction, a distinction that binary metrics alone would fail to capture. Second, **statistical priors are effective teachers**: injecting annotation-derived timing statistics directly into prompts provides a simple but powerful mechanism for improving temporal coherence without any fine-tuning. Third, **personalization is hard**: persona scores improve only marginally, suggesting that aligning LLM-generated itineraries with nuanced traveler preferences requires richer mechanisms, potentially including multi-turn dialogue or retrieval-augmented persona grounding.

The low HCPR and Final Pass Rates across both settings confirm that *TripCraft* presents a substantially more challenging evaluation environment

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Setting	Category	$\bar{T}_{meal}$	$\bar{T}_{attrac}$	$\bar{S}_{spatial}$	$\bar{S}_{persona}$	$\bar{S}_{ord}$
w/o Parameter Info	3-day	0.56	0.0169	0.80	0.46	0.70
	5-day	0.69	0.0057	0.86	0.49	0.91
	7-day	0.61	**0.0122**	0.85	0.50	0.96
w/ Parameter Info	3-day	**0.70**	**0.0171**	**0.83**	**0.48**	**0.74**
	5-day	**0.75**	**0.0078**	**0.87**	**0.50**	**0.93**
	7-day	**0.80**	0.0057	**0.86**	**0.51**	**0.97**

Table 6: Results of our proposed evaluation metrics for the two settings. For each category, the **bolded** values represent the better ones between the two settings.

Setting	Category	Delivery Rate	CPR		HCPR		Final Pass Rate
			Micro	Macro	Micro	Macro	
w/o Parameter Info	3-day	92.60	77.61	**6.08**	**27.63**	**26.08**	**3.47**
	5-day	**98.69**	**66.99**	**2.17**	4.21	3.04	**1.74**
	7-day	**96.03**	**70.35**	0.00	0.62	0.00	0.00
w/ Parameter Info	3-day	**96.08**	**80.08**	5.21	26.10	25.21	1.74
	5-day	91.30	61.69	0.87	**8.60**	**5.21**	0.43
	7-day	92.51	66.43	0.00	**0.83**	0.00	0.00

Table 7: Results of existing evaluation metrics for the two settings. CPR and HCPR stand for Commonsense Pass Rate and Hard Constraint Pass Rate respectively. For each category, the **bolded** values represent the better ones between the two settings. Moreover, equal values have been underlined.

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than prior benchmarks, making it a useful testbed for future advances in LLM-based planning (Hao et al., 2025; Lee et al., 2025).

A closer analysis of constraint failures reveals that the most common violations in the *w/ Parameter Info* setting occur in meal gap constraints: when the model adheres tightly to annotation-derived meal windows (e.g., scheduling lunch between 11:30 and 13:00), it occasionally produces itineraries where the mandated four-hour inter-meal gap is violated, particularly in 5-day trips with dense attraction schedules. In contrast, the baseline model, unconstrained by timing priors, more freely distributes meals across the day, achieving gap compliance at the cost of temporal coherence. This pattern suggests that future work should explore *soft constraint integration*, where statistical timing priors are incorporated as soft penalties within a planning objective rather than as rigid prompt instructions. Such an approach could resolve the tension between temporal realism and hard constraint satisfaction, enabling models to benefit from annotation-derived priors without sacrificing feasibility. Similarly, per-city attraction density varies considerably across the 140 cities in *TripCraft*, and models often struggle to select transit-proximate PoIs in sparser cities, contributing to lower spatial scores in hard-difficulty queries.

5 Conclusion

We presented *TripCraft*, a spatio-temporally fine-grained travel planning benchmark with 1,000 queries across 140 U.S. cities. *TripCraft* integrates real-world data from public transit feeds, event calendars, flight records, and curated PoI databases to create a constraint-rich evaluation environment. A distinctive feature of *TripCraft* is its suite of five continuous metrics—Temporal Meal, Temporal Attraction, Spatial, Ordering, and Persona scores—whose parameters are calibrated from human-annotated gold itineraries, enabling nuanced assessment of itinerary quality beyond binary pass rates.

Experiments with GPT-4o demonstrate that parameter-informed prompting substantially improves temporal scheduling quality, with 7-day temporal meal scores rising from 0.61 to 0.80, while also revealing a rigidity-versus-feasibility trade-off that binary constraint metrics alone cannot surface. These findings highlight both the promise of statistics-guided prompting and the substantial headroom remaining for improving LLM-based travel planning.

We release *TripCraft* as a public benchmark to encourage future work on realistic, personalized, and transit-aware itinerary generation. Key directions for future research include multi-turn plan-

ning with persona refinement, integration of dynamic constraint solvers, and fine-grained annotation of feasibility rationales.

References

Bernd Bohnet, Azade Nova, Aaron T Parisi, Kevin Swersky, Katayoon Goshvadi, Hanjun Dai, Dale Schuurmans, Noah Fiedel, and Hanie Sedghi. 2024. Exploring and benchmarking the planning capabilities of large language models. arXiv preprint arXiv:2406.13094.

Andrey Borro, Patricia J Riddle, Michael W Barley, and Michael J Witbrock. 2025. Large language models as common-sense heuristics. arXiv preprint arXiv:2501.18816.

Aili Chen, Xuyang Ge, Ziquan Fu, Yanghua Xiao, and Jiangjie Chen. 2024. Travelagent: An ai assistant for personalized travel planning. Preprint, arXiv:2409.08069.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.

Atharva Gundawar, Mudit Verma, Lin Guan, Karthik Valmeekam, Siddhant Bhambri, and Subbarao Kambhampati. 2024. Robust planning with llm-modulo framework: Case study in travel planning. arXiv preprint arXiv:2405.20625.

Yilun Hao, Yongchao Chen, Yang Zhang, and Chuchu Fan. 2025. Large language models can solve real-world planning rigorously with formal verification tools. Preprint, arXiv:2404.11891.

Zhijian Huang, Tao Tang, Shaoxiang Chen, Sihao Lin, Zequn Jie, Lin Ma, Guangrun Wang, and Xiaodan Liang. 2024. Making large language models better planners with reasoning-decision alignment. In European Conference on Computer Vision, pages 73–90. Springer.

Ramya Jonnala, Gongbo Liang, Jeong Yang, and Izzat Alsmadi. 2025. Exploring the potential of large language models in public transportation: San antonio case study. arXiv preprint arXiv:2501.03904.

Kuang-Huei Lee, Ian Fischer, Yueh-Hua Wu, Dave Marwood, Shumeet Baluja, Dale Schuurmans, and Xinyun Chen. 2025. Evolving deeper llm thinking. arXiv preprint arXiv:2501.09891.

Vladimir I. Levenshtein. 1965. Binary codes capable of correcting deletions, insertions, and reversals. Soviet physics. Doklady, 10:707–710.

Vishal Pallagani, Bharath Muppasani, Keerthiram Murugesan, Francesca Rossi, Biplav Srivastava, Lior Horesh, Francesco Fabiano, and Andrea Loreggia. 2023. Understanding the capabilities of large language models for automated planning. arXiv preprint arXiv:2305.16151.

Archiki Prasad, Alexander Koller, Mareike Hartmann, Peter Clark, Ashish Sabharwal, Mohit Bansal, and Tushar Khot. 2024. Adapt: As-needed decomposition and planning with language models. In Findings of the Association for Computational Linguistics: NAACL 2024, pages 4226–4252.

Jie-Jing Shao, Xiao-Wen Yang, Bo-Wen Zhang, Baizhi Chen, Wen-Da Wei, Lan-Zhe Guo, and Yu-feng Li. 2024. Chinatravel: A real-world benchmark for language agents in chinese travel planning. arXiv preprint arXiv:2412.13682.

Harmanpreet Singh, Nikhil Verma, Yixiao Wang, Manasa Bharadwaj, Homa Fashandi, Kevin Ferreira, and Chul Lee. 2024. Personal large language model agents: A case study on tailored travel planning. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track, pages 486–514, Miami, Florida, US. Association for Computational Linguistics.

Chan Hee Song, Jiaman Wu, Clayton Washington, Brian M. Sadler, Wei-Lun Chao, and Yu Su. 2023. Llm-planner: Few-shot grounded planning for embodied agents with large language models. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV).

Karthik Valmeekam, Sarath Sreedharan, Matthew Marquez, Alberto Olmo, and Subbarao Kambhampati. 2023. On the planning abilities of large language models (a critical investigation with a proposed benchmark). Preprint, arXiv:2302.06706.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. 2022. Chain-of-thought prompting elicits reasoning in large language models. Advances in neural information processing systems, 35:24824–24837.

Zhiheng Xi, Wenxiang Chen, Xin Guo, Wei He, Yiwen Ding, Boyang Hong, Ming Zhang, Junzhe Wang, Senjie Jin, Enyu Zhou, et al. 2025. The rise and potential of large language model based agents: A survey. Science China Information Sciences, 68(2):121101.

Jian Xie, Kai Zhang, Jiangjie Chen, Tinghui Zhu, Renze Lou, Yuandong Tian, Yanghua Xiao, and Yu Su. 2024. Travelplanner: A benchmark for real-world planning with language agents. In Forty-first International Conference on Machine Learning.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. React: Synergizing reasoning and acting in language models. Preprint, arXiv:2210.03629.

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Zirui Zhao, Wee Sun Lee, and David Hsu. 2024. Large language models as commonsense knowledge for large-scale task planning. Advances in Neural Information Processing Systems, 36.

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